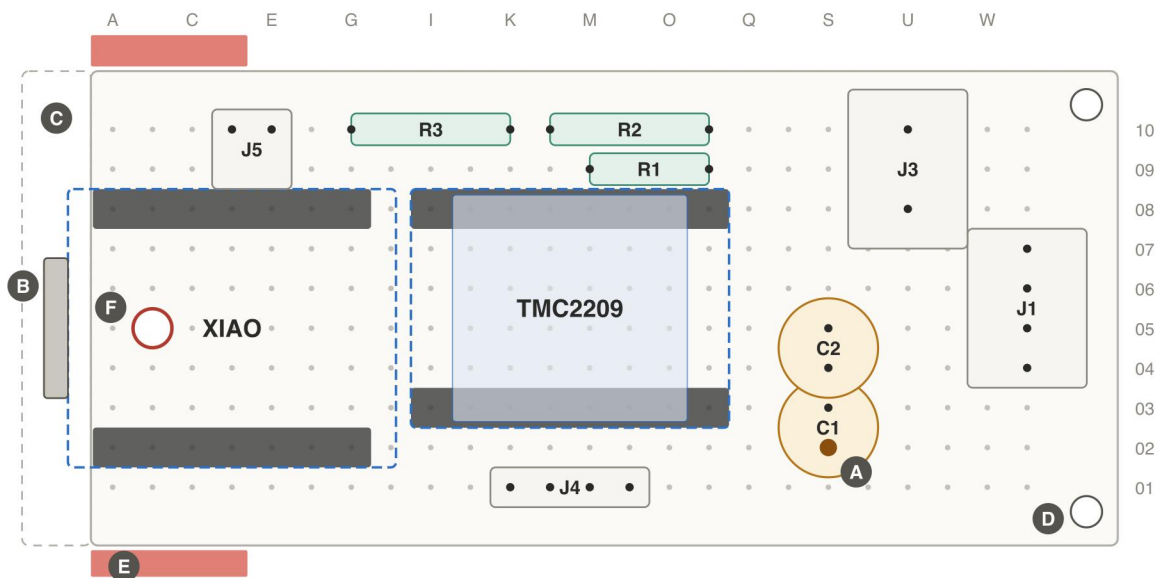


Board · where each component goes

1 · where each component goes



- 30 × 70 perfboard, 10 × 24 holes

XIAO and driver boards, on the sockets

electrolytics (C1, C2)

resistors, lying flat
- female sockets, with the rows of pins

driver heatsink

terminal blocks and headers

panel guides

Components

- C1

47 μ F 63 V, driver VM

C2

47 μ F 63 V, 12 V input (optional)

J1

phase A and B terminal block, four poles

J3

12 V terminal block

J4

sensor header, 4 poles

J5

LED header, 2 poles

R1

10 k Ω , EN pull-up

R2

1 k Ω in series, UART

R3

1 k Ω in series, LED

pins in S03, S02

pins in S05, S04

pins in X07, X06, X05, X04

pins in U10, U08

pins in K01, L01, M01, N01

pins in D10, E10

pins in M09, P09

pins in L10, P10

pins in G10, K10

Points

- A

single ground node: the negative of C1, in S02. One wire per branch from here.

B

XIAO USB socket: flush with the outer face of the head.

C

part removed by the cut: the board ends 1.38 mm past column C0.

D

fixing holes: only at the far end; those at the head go with the cut.

E

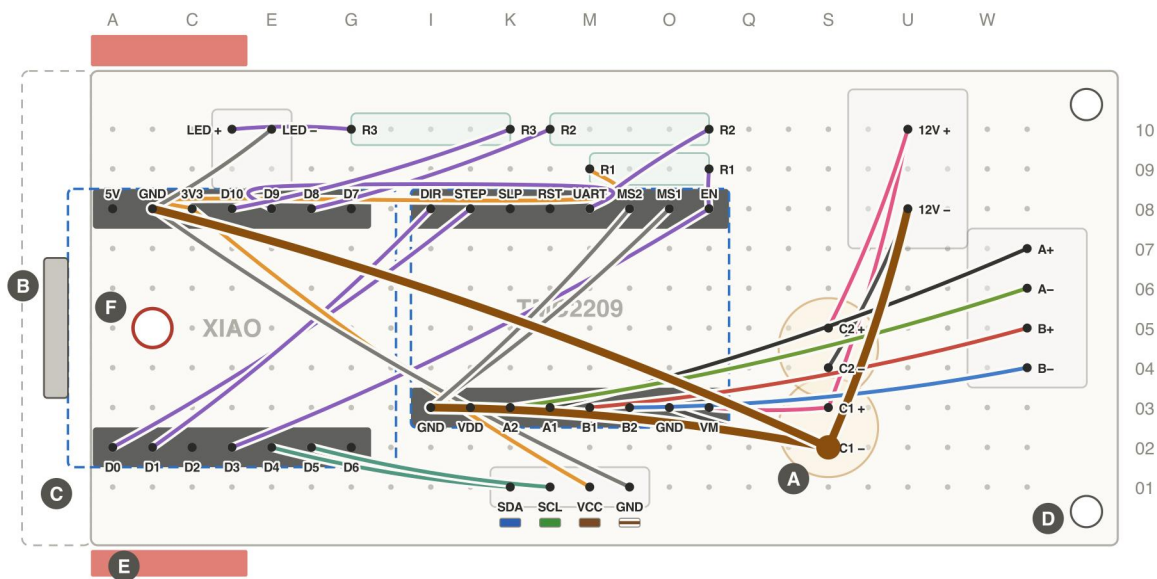
panel guides: they square the board from the side, they do not cover it. It goes in from above.

F

hole B05, to drill out to $\varnothing$ 2.7: the third screw, the one that holds the cut end.

Board · the wires seen from above and from below

2 · The wires, with the star ground

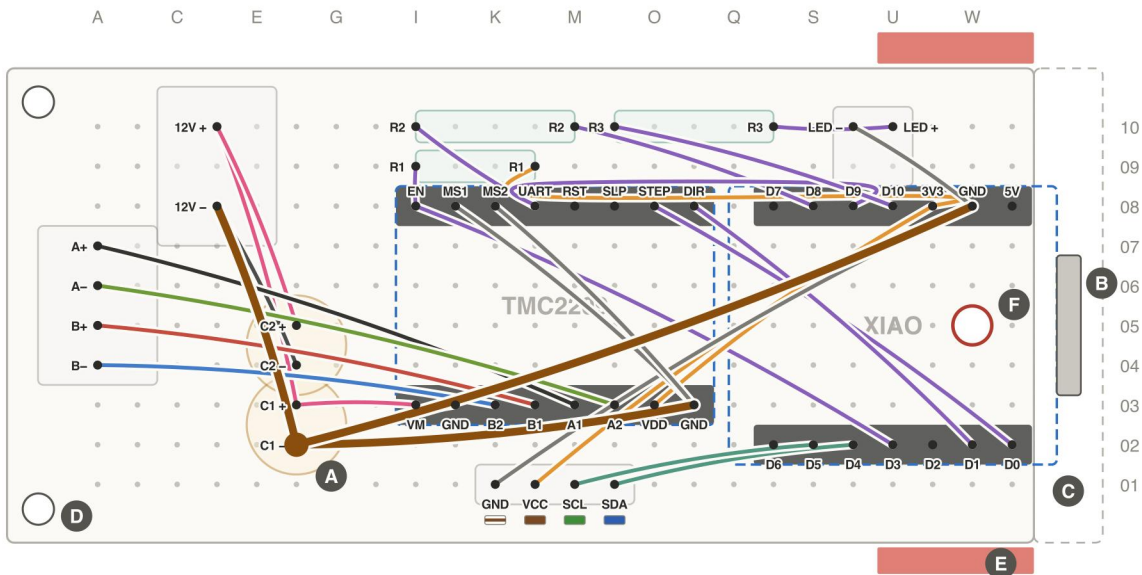


Nets

- 12 V and VM
- power ground: lead of C1, - of C2
- XIAO 3V3
- sensor I2C
- phase A- (green)
- phase B- (blue)
- at J4: the colour of the sensor cable (CAT5) wire on each pin - white-brown GND, brown VCC, green SCL, blue SDA
- star ground: the wires leaving the node
- logic ground, local bus of its branch
- signals
- phase A+ (black)
- phase B+ (red)

3 · The same board seen FROM BELOW, on the solder side

The drawing is flipped the way the board is flipped; the letters still run left to right, because they are printed on this face too.



WARNING: the orientation of the XIAO (D0–D6 on row R8) comes from the maker's pinout turned with the USB to the left; the old layout had it the other way round. Check it on the real XIAO before soldering. Positions and wires come from mechanics/wheelly-cad/src/electronics.py and wiring.py: moving a component there moves the check that it clears the box as well. The bodies of terminal blocks, headers and resistors are catalogue ESTIMATES. The wires are drawn smooth, with a white halo at crossings to show which is on top. They say what goes where, not the route.

Board · the star ground and the list of wires

The star ground, as it is done here

The node is the negative of C1, in S02: its lead goes short to the driver power GND (O03).

THREE wires leave the node, and only three: to – of the 12 V block, to the driver logic GND (I02), to the XIAO GND (B08).

Each branch then has its own local bus, which touches no other: from the XIAO the sensor and LED ground,

from the driver logic GND, MS1 and MS2. No ground connects to another ground outside the node.

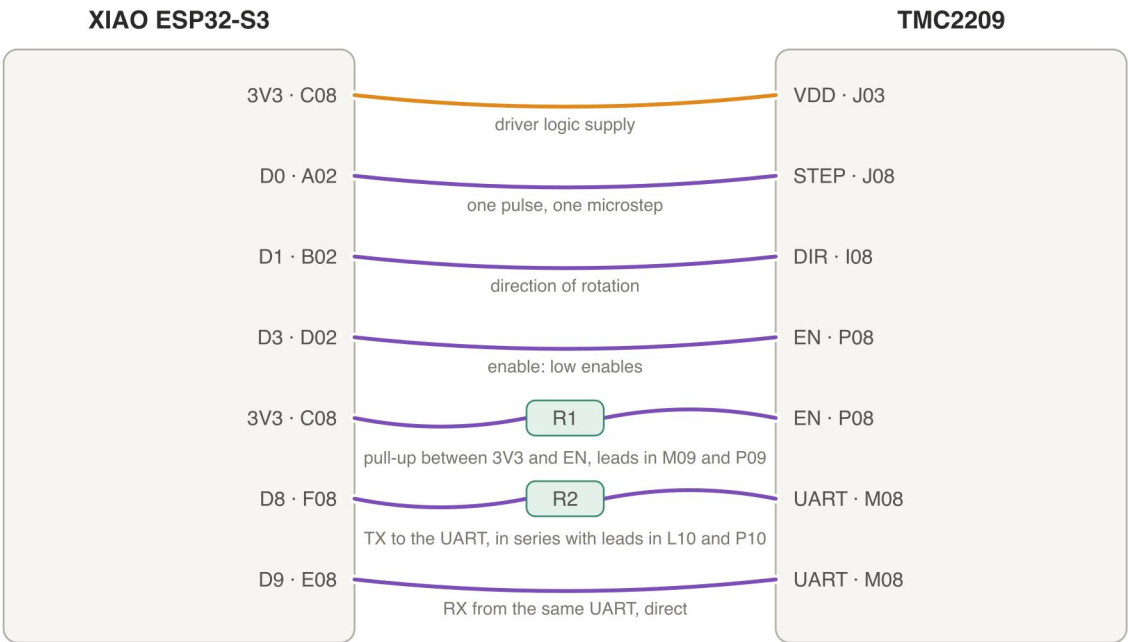
C2, if fitted, sits on the pins of the block: its – goes to – of the block, not to the node.

Wires

In brackets the hole: first as read FROM ABOVE, on the component side, then as read FROM BELOW, on the solder side.

— driver VM (P03/I03) → C1 + (S03/F03)	— R1 (P09/I09) → driver EN (P08/I08)
— driver GND (O03/J03) → C1 – (S02/F02)	— R1 (M09/L09) → XIAO 3V3 (C08/V08)
— C1 + (S03/F03) → 12V + (U10/D10)	— XIAO 3V3 (C08/V08) → driver VDD (J03/O03)
— C2 + (S05/F05) → 12V + (U10/D10)	— XIAO 3V3 (C08/V08) → VCC (M01/L01)
— C2 – (S04/F04) → 12V – (U08/D08)	— XIAO D8 (F08/S08) → R2 (L10/M10)
— C1 – (S02/F02) → 12V – (U08/D08)	— R2 (P10/I10) → driver UART (M08/L08)
— C1 – (S02/F02) → driver GND (I03/P03)	— XIAO D9 (E08/T08) → driver UART (M08/L08)
— C1 – (S02/F02) → XIAO GND (B08/W08)	— XIAO D4 (E02/T02) → SDA (K01/N01)
— driver A1 (L03/M03) → A+ (X07/A07)	— XIAO D5 (F02/S02) → SCL (L01/M01)
— driver A2 (K03/N03) → A– (X06/A06)	— XIAO D10 (D08/U08) → R3 (K10/N10)
— driver B1 (M03/L03) → B+ (X05/A05)	— R3 (G10/R10) → LED + (D10/U10)
— driver B2 (N03/K03) → B– (X04/A04)	— XIAO GND (B08/W08) → LED – (E10/T10)
— XIAO D0 (A02/X02) → driver STEP (J08/O08)	— XIAO GND (B08/W08) → GND (N01/K01)
— XIAO D1 (B02/W02) → driver DIR (I08/P08)	— driver GND (I03/P03) → driver MS1 (O08/J08)
— XIAO D3 (D02/U02) → driver EN (P08/I08)	— driver GND (I03/P03) → driver MS2 (N08/K08)

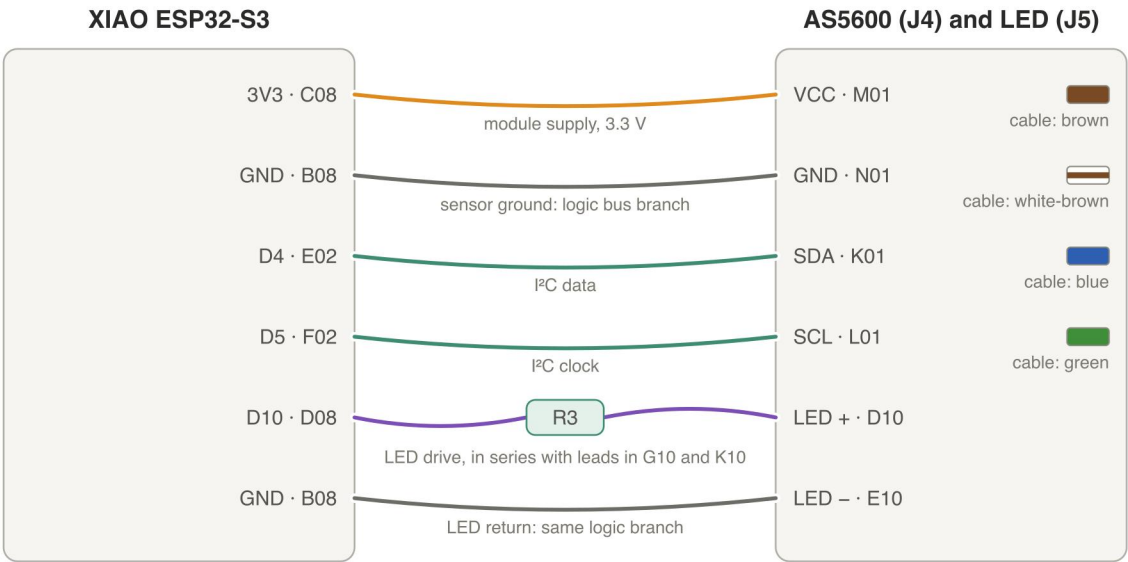
Connections · driver, sensor and LED



THE GROUND DOES NOT GO THROUGH HERE. The driver GND (I02) and the XIAO GND (B08) both leave the node, the negative of C1 (S02), with one wire each.

Joining them would close the loop that pcb/README says not to close.

The driver's MS1 and MS2 go to its logic ground, inside its own branch.

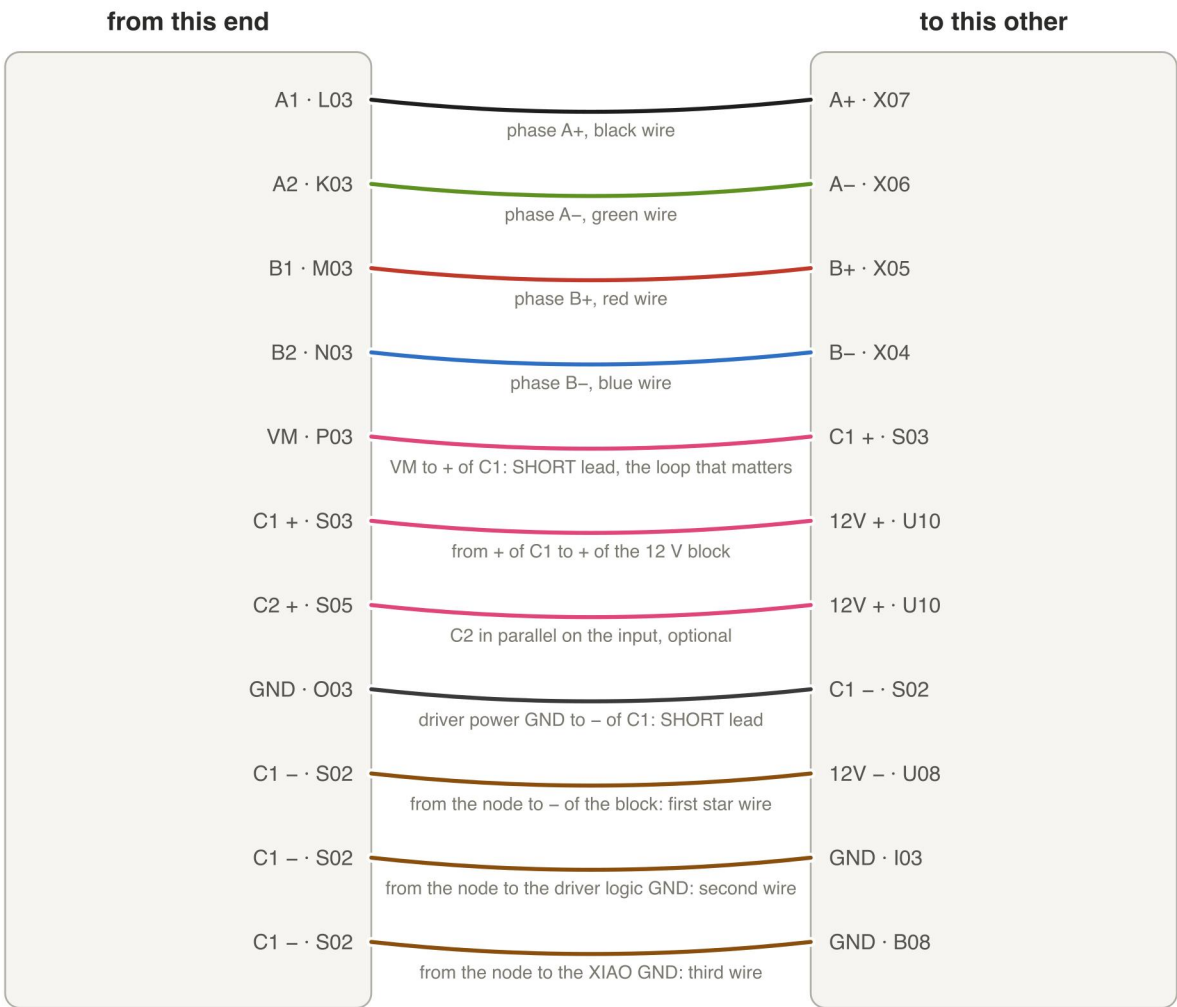


These two grounds leave the XIAO GND and not the node: they are its branch of the logic bus, and inside a branch the grounds may touch.

The module's 104 capacitor sits under its own board, not here.

SENSOR CABLE (CAT5), four wires, colours as soldered on the module: brown 3V3, white-brown GND, blue SDA, green SCL. DIR is tied to GND on the module (jumper pad 2 - pad 1), not through the cable. Cut back and insulated at this end: white-green, white-blue, orange, white-orange.

Connections · power, phases and ground node



THE GROUND NODE IS THE NEGATIVE OF C1, IN S02, and THREE WIRES AND ONLY THREE leave it.

No branch connects to another branch: two grounds touching in two places close a loop, and part of the motor return runs along the USB shield to the camera.

The two leads of C1 are cut short, straddling the VM and GND pins of the socket: the area of that loop counts more than any shielding downstream, and comes before the space it takes. That is why C1 stayed here and C2 moved.

The phase colours are the motor's, not a choice: black and green one phase, red and blue the other. Swapping the two wires of ONE phase reverses the rotation.